

Solution of MGMT 670 Homework 2

Due 11:59 pm EST on September 15. Submit via Brightspace.

1. Susan, the marketing manager of a consumer electronics company, is considering whether to introduce a new large screen television model into the market. Susan is aware that the success of the new model depends on the economy in the coming year which in turn depends on the result of the upcoming presidential election. According to her assessment, the incumbent and the challenger are equally likely to win the elections. A fixed cost of 3 million dollars has been incurred prior to making the decision to launch the model. The payoff table (in millions) for the two alternative actions is set up as follows (the fixed cost has been included in the calculation):

Introduction of the new model	Presidential Election Outcome	
	Incumbent	Challenger
Yes	-35	45
No	-3	-3

- (a) Find the expected payoffs of the two alternatives. Which is better?
- (b) Bob, a well known consultant, proposed a project for Susan to consider. Bob argued that he could provide a prediction of the election result through a special survey method. Susan just graduated and received an A grade in MGMT 670 two years ago. She still remembers the correct method of using Bayes' Theorem. After interviewing Bob, Susan obtained the following probability estimates:

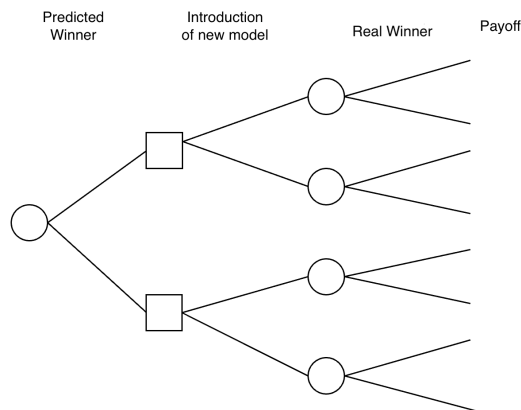
$$P(\text{Predicted Winner is Incumbent} \mid \text{Real Winner is Incumbent}) = .8$$

$$P(\text{Predicted Winner is Challenger} \mid \text{Real Winner is Incumbent}) = .2$$

$$P(\text{Predicted Winner is Incumbent} \mid \text{Real Winner is Challenger}) = .3$$

$$P(\text{Predicted Winner is Challenger} \mid \text{Real Winner is Challenger}) = .7.$$

Complete the following decision tree for evaluating the expected payoff with the prediction result (sample information) from Bob. Compute EVSI.



Solution:

- (a) EV of introducing the new model = $(-35)(.5) + (45)(.5) = 5$.
 EV of not introducing the new model = $(-3)(.5) + (-3)(.5) = -3$.
 Therefore, introducing the new model gives a higher expected payoff.
- (b) $P(\text{Predicted Winner} = \text{Incumbent}) = P(\text{Predicted Winner} = \text{Incumbent} \mid \text{Real Winner} = \text{Incumbent}) \cdot P(\text{Real Winner} = \text{Incumbent}) + P(\text{Predicted Winner} = \text{Incumbent} \mid \text{Real Winner} = \text{Challenger}) \cdot P(\text{Real Winner} = \text{Challenger}) = 0.8 \cdot 0.5 + 0.3 \cdot 0.5 = 0.55$.
 $P(\text{Predicted Winner} = \text{Challenger}) = 1 - 0.55 = 0.45$.
 $P(\text{Real Winner} = \text{Incumbent} \mid \text{Predicted Winner} = \text{Incumbent}) = P(\text{Predicted Winner} = \text{Incumbent} \mid \text{Real Winner} = \text{Incumbent}) \cdot P(\text{Real Winner} = \text{Incumbent}) / P(\text{Predicted Winner} = \text{Incumbent}) = 0.8 \cdot 0.5 / 0.55 = 0.727$
 $P(\text{Real Winner} = \text{Challenger} \mid \text{Predicted Winner} = \text{Incumbent}) = 1 - 0.727 = 0.273$
 $P(\text{Real Winner} = \text{Challenger} \mid \text{Predicted Winner} = \text{Challenger}) = 0.7 \cdot 0.5 / 0.45 = 0.778$
 $P(\text{Real Winner} = \text{Incumbent} \mid \text{Predicted Winner} = \text{Challenger}) = 1 - 0.778 = 0.222$.
 EV with sample information = $0.55 \cdot (-3) + 27.24 \cdot 0.45 = 10.608$
 So EVSI = $10.608 - 5 = 5.608$.

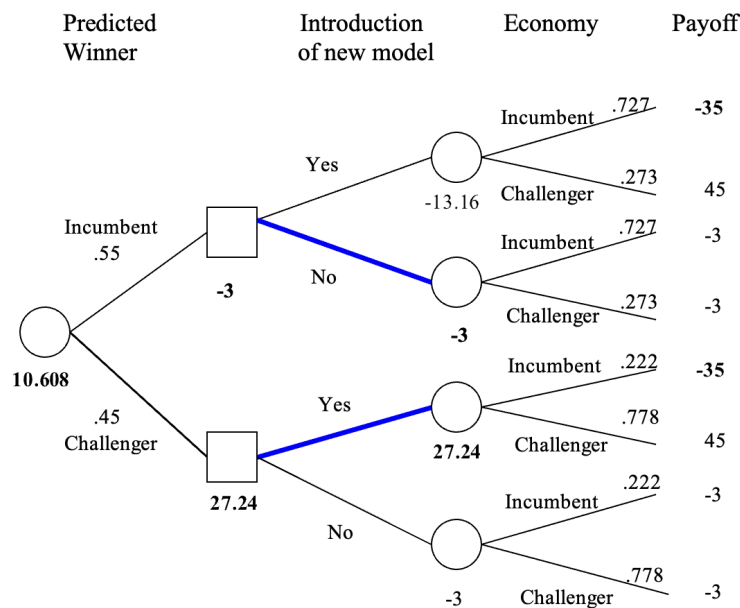


Figure 1: Solution of Q1: Decision tree.

2. Over the past 10 years, the following statistics were observed for three investment options:

Investment	Expected Return	Standard Deviation
S&P 500	12.57%	15.12%
Core Bonds	2.12%	2.13%
Real Estate Investment Trusts (REITs)	5.47%	9.00%

The correlations between the investments are:

$$\rho_{\text{S\&P, Bonds}} = -0.32, \quad \rho_{\text{S\&P, REITs}} = 0.74, \quad \rho_{\text{Bonds, REITs}} = -0.04$$

- (a) Construct a portfolio that is 50% in S&P 500 and 50% in REITs. Compute the expected return and standard deviation of this portfolio.
- (b) Construct a portfolio that is 50% in Core Bonds and 50% in REITs. Compute the expected return and standard deviation of this portfolio.
- (c) Construct a portfolio that is 80% in Core Bonds and 20% in REITs. Compute the expected return and standard deviation of this portfolio.
- (d) Based on your results, which portfolio would you recommend to an aggressive investor? Which would you recommend to a conservative investor? Justify your answer.

Solution:

Given Data:

- S&P 500: $E[R_S] = 12.57\%$, $\sigma_S = 15.12\%$
- Core Bonds: $E[R_B] = 2.12\%$, $\sigma_B = 2.13\%$
- REITs: $E[R_R] = 5.47\%$, $\sigma_R = 9.00\%$

Correlations:

$$\rho_{S,B} = -0.32, \quad \rho_{S,R} = 0.74, \quad \rho_{B,R} = -0.04$$

1. Portfolio: 50% S&P, 50% REITs

Expected Return:

$$E[R_p] = 0.5 \cdot 12.57\% + 0.5 \cdot 5.47\% = 9.02\%$$

Variance:

$$\sigma_p^2 = (0.5)^2 \cdot (15.12\%)^2 + (0.5)^2 \cdot (9.00\%)^2 + 2 \cdot 0.5 \cdot 0.5 \cdot 0.74 \cdot 15.12\% \cdot 9.00\% \approx 0.01278$$

$$\sigma_p = \sqrt{0.01278} \approx 11.30\%$$

2. Portfolio: 50% Bonds, 50% REITs

Expected Return:

$$E[R_p] = 0.5 \cdot 2.12\% + 0.5 \cdot 5.47\% = 3.80\%$$

Variance:

$$\sigma_p^2 = (0.5)^2 \cdot (2.13\%)^2 + (0.5)^2 \cdot (9.00\%)^2 + 2 \cdot 0.5 \cdot 0.5 \cdot (-0.04) \cdot 2.13\% \cdot 9.00\% \approx 0.0021$$

$$\sigma_p = \sqrt{0.0021} \approx 4.58\%$$

3. Portfolio: 80% Bonds, 20% REITs

Expected Return:

$$E[R_p] = 0.8 \cdot 2.12\% + 0.2 \cdot 5.47\% = 2.79\%$$

Variance:

$$\sigma_p^2 = (0.8)^2 \cdot (2.13\%)^2 + (0.2)^2 \cdot (9.00\%)^2 + 2 \cdot 0.8 \cdot 0.2 \cdot (-0.04) \cdot 2.13\% \cdot 9.00\% \approx 0.0005898$$

$$\sigma_p = \sqrt{0.0005898} \approx 2.43\%$$

4. Recommendations:

- **Aggressive Investor:** Prefers higher returns. Portfolio with 50% S&P and 50% REITs has the highest expected return with moderate risk. **Recommended portfolio: 50% S&P, 50% REITs.**
 - **Conservative Investor:** Prefers low risk. Portfolio with 80% Bonds and 20% REITs has the lowest risk with modest return. **Recommended portfolio: 80% Bonds, 20% REITs.**
3. According to a financial study, 18% of retail investors hold cryptocurrency in their investment portfolios. Consider a randomly selected group of 25 retail investors.
- (a) Compute the probability that exactly 7 investors hold cryptocurrency in their portfolios.
 - (b) Compute the probability that at least 10 of the investors hold cryptocurrency in their portfolios.
 - (c) Suppose you observed that 12 investors in the group hold cryptocurrency. Would this observation make you question the accuracy of the 18% figure? Explain your reasoning.
 - (d) If the first 5 investors in the sample are known to all hold cryptocurrency, what is the probability that exactly 7 investors in total hold cryptocurrency in the group of 25?

Solution: Given:

- Probability an investor holds cryptocurrency: $p = 0.18$
- Number of investors: $n = 25$
- Random variable $X \sim \text{Binomial}(n = 25, p = 0.18)$

(a) Probability that exactly 7 investors hold cryptocurrency:

$$P(X = 7) = \binom{25}{7} (0.18)^7 (0.82)^{18}$$

Using Excel:

$$= \text{BINOM.DIST}(7, 25, 0.18, \text{FALSE}) = 0.0827$$

$$P(X = 7) \approx 0.0827$$

(b) Probability that at least 10 investors hold cryptocurrency:

$$P(X \geq 10) = 1 - P(X \leq 9)$$

Using Excel:

$$= 1 - \text{BINOM.DIST}(9, 25, 0.18, \text{TRUE}) = 0.00831$$

$$P(X \geq 10) \approx 0.00831$$

(c) Observing 12 investors with cryptocurrency:

- 12 out of 25 = 48% is much higher than 18%. - Probability of 12 or more investors:

$$P(X \geq 12) = 1 - P(X \leq 11)$$

Using Excel:

$$= 1 - \text{BINOM.DIST}(11, 25, 0.18, \text{TRUE}) = 0.000578$$

Extremely unlikely. Thus, this observation strongly suggests that the 18% survey may be inaccurate.

(d) Conditional probability (first 5 investors known to hold cryptocurrency):

- Let Y = number of cryptocurrency investors in remaining 20 investors.
- Total exactly 7 investors, which means 2 more successes needed in remaining 20:

$$Y \sim \text{Binomial}(n = 20, p = 0.18), \quad \text{compute } P(Y = 2)$$

Using Excel:

$$= \text{BINOM.DIST}(2, 20, 0.18, \text{FALSE}) = 0.173$$

$$P(\text{exactly 7 total} \mid 5 \text{ first}) \approx 0.173$$

4. A manufacturer of electric vehicle (EV) batteries claims that the lifetime of its lithiumion batteries (before capacity drops below 80%) is normally distributed. From internal testing data, the average lifetime is 4500 charge cycles with a standard deviation of 420 cycles.

- The company wants to offer a warranty for the top 90% of its batteries. What is the minimum number of charge cycles a battery must last to qualify for this warranty?
- A customer purchases an EV battery and manages to achieve 5200 charge cycles before reaching 80% capacity. What is the probability that a randomly selected battery lasts at least this long?
- The company wants to ensure that only 5% of batteries fail before the guaranteed lifetime. What should be the minimum lifetime guaranteed to achieve this goal?
- Suppose the batteries are tested in batches of 50 units. Approximately how many batteries in a batch would you expect to fail before 4000 charge cycles?

Solution:

- Battery lifetime $X \sim N(\mu = 4500, \sigma = 420)$ charge cycles

(a) Minimum lifetime for top 90% warranty:

Top 90% means 90% of batteries last at least that many cycles. So we need the 10th percentile of the normal distribution.

- Find $Z_{0.10}$ from the standard normal table: $Z_{0.10} \approx -1.285$.
- Convert to battery cycles:

$$X = \mu + Z \cdot \sigma = 4500 - 1.285 \cdot 420 \approx 3960 \text{ cycles}$$

It's OK if students use $Z_{0.10} \approx -1.28$, which leads to the answer 3962.

(b) Probability a battery lasts at least 5200 cycles:

- Find z-score:

$$Z = \frac{X - \mu}{\sigma} = \frac{5200 - 4500}{420} \approx \frac{700}{420} \approx 1.667$$

- Probability:

$$P(X \geq 5200) = 1 - P(Z \leq 1.667) \approx 1 - 0.9525 = 0.0475$$

$$P(X \geq 5200) \approx 4.75\%$$

(c) Lifetime guaranteeing that only 5% fail before a certain number of cycles:

- We want $P(X < X_{\min}) = 0.05 \rightarrow$ find $Z_{0.05} \approx -1.645$
- Convert to battery cycles:

$$X_{\min} = \mu + Z \cdot \sigma = 4500 + (-1.645) \cdot 420 \approx 4500 - 690.9 \approx 3809 \text{ cycles}$$

$$X_{\min} \approx 3809 \text{ cycles}$$

(d) Expected number of batteries failing before 4000 cycles in a batch of 50:

- Find z-score for 4000 cycles:

$$Z = \frac{4000 - 4500}{420} = \frac{-500}{420} \approx -1.19$$

- Probability:

$$P(X < 4000) = P(Z < -1.19) \approx 0.117$$

- Expected number in batch of 50:

$$E[\text{failures}] = 50 \cdot 0.117 \approx 5.85 \approx 6 \text{ batteries}$$

Expected failures: approximately 6 batteries
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